

MERIDA INTL, Mexico

WMO: 766440

Lat: 20.9469N

Lon: 89.6522W

Elev: 36

StdP: 14.68

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.8	60.2	49.6	52.8	71.5	51.7	57.1	71.8	22.7	79.7	20.9	80.1	2.2	110	0.387

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	23.4	101.8	76.0	99.4	76.1	97.6	76.2	84.1	90.7	82.6	89.2	81.4	88.4	10.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	170.4	87.0	81.0	161.0	85.1	80.2	157.1	84.4	48.3	90.3	46.5	87.2	45.2	89.0	94.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.7	18.9	17.1	DB	48.0	105.1	2.7	2.3	46.1	106.8	44.5	108.2	43.0	109.5	41.1	111.2	
			WB	47.2	86.7	3.0	2.6	45.0	88.6	43.3	90.1	41.6	91.6	39.4	93.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	74.9	77.7	80.5	84.2	86.4	85.1	84.4	84.4	83.8	81.2	77.6	76.0
	DBStd	5.35	4.97	5.07	5.15	4.29	3.22	2.71	1.96	1.85	2.24	3.30	4.04	4.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	2	1	1	0	0	0	0	0	0	0	0	0	0
	CDD50	11444	773	775	944	1025	1127	1052	1066	1067	1013	969	828	805
	CDD65	5971	309	356	479	575	662	602	601	602	563	504	378	340
	CDH74	60403	2336	3173	4888	6612	7992	6606	6322	6320	5730	4752	3109	2563
	CDH80	28751	909	1465	2512	3633	4410	3254	3039	3003	2603	1889	1099	935
Wind	WSAvg	7.6	7.3	8.4	9.7	10.1	9.6	8.3	7.1	6.3	5.8	6.2	6.3	6.5
Precipitation	PrecAvg	40.1	1.3	1.1	1.0	1.3	2.4	6.3	5.8	6.1	6.7	4.6	1.9	1.6
	PrecMax	50.2	5.2	5.9	2.5	5.0	5.0	12.0	9.2	14.6	12.0	11.7	5.9	5.1
	PrecMin	26.3	0.0	0.1	0.1	0.0	0.7	1.5	2.0	2.5	2.1	1.0	0.4	0.2
	PrecStd	5.8	1.4	1.3	0.6	1.3	1.4	2.4	1.7	2.4	2.6	2.4	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	97.1	100.8	104.3	104.4	101.1	98.9	98.8	99.0	96.6	93.4	93.2
		MCWB	73.4	72.8	74.4	75.5	75.9	78.0	78.0	79.8	80.4	79.2	77.0	75.5
	2%	DB	91.2	94.5	98.2	100.7	102.1	98.6	96.7	96.5	95.4	93.2	91.1	90.0
		MCWB	73.1	73.2	73.9	75.0	75.8	78.0	79.0	79.6	79.8	79.2	76.3	74.5
	5%	DB	88.6	91.7	95.4	98.6	99.7	96.5	94.6	94.4	93.3	91.3	88.4	88.0
		MCWB	72.7	72.5	73.1	74.4	75.8	78.2	78.9	79.4	79.8	79.0	75.7	74.2
	10%	DB	85.7	89.4	92.9	96.4	96.9	93.4	92.8	92.7	91.5	89.4	86.2	85.8
		MCWB	72.1	72.3	72.6	74.1	76.3	78.2	78.6	79.2	79.4	78.1	74.7	73.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	79.3	80.2	82.0	83.6	84.5	84.9	85.5	85.7	84.2	81.9	81.0
		MCDB	84.8	88.8	90.2	92.2	93.7	91.5	91.7	92.5	92.2	89.4	86.5	86.4
	2%	WB	76.4	76.7	77.4	79.2	80.8	82.2	82.6	83.1	83.3	81.8	79.4	78.0
		MCDB	84.4	85.9	89.2	90.4	90.4	89.1	89.0	89.7	89.6	87.6	85.4	84.6
	5%	WB	75.1	75.3	76.0	77.8	79.3	80.8	81.0	81.6	81.7	80.5	78.2	76.4
		MCDB	83.2	84.5	87.4	89.6	89.9	88.0	88.0	88.7	88.2	86.9	84.5	83.6
	10%	WB	73.8	74.2	74.9	76.7	78.3	79.8	80.0	80.5	80.5	79.4	76.9	75.2
		MCDB	80.9	82.8	85.6	88.6	89.4	87.6	87.6	88.0	87.4	85.9	83.3	81.6
Mean Daily Temperature Range	5% DB	MDBR	21.1	22.9	24.4	24.9	23.4	19.9	20.0	19.8	18.9	17.6	19.0	19.9
		MCDBR	24.2	25.7	27.3	27.6	26.3	22.7	21.7	21.4	20.8	19.2	20.4	21.9
	5% WB	MCWBR	10.0	9.2	8.8	8.0	7.8	8.1	9.1	9.3	9.1	8.5	9.3	9.8
		MCWBR	20.9	22.6	24.7	24.6	23.3	20.0	20.2	20.1	19.2	17.6	17.6	19.1
Clear-Sky Solar Irradiance	taub		0.358	0.372	0.400	0.435	0.451	0.448	0.465	0.441	0.411	0.395	0.375	0.371
		taud	2.554	2.509	2.402	2.292	2.292	2.348	2.283	2.366	2.439	2.486	2.505	2.521
	Ebn at Noon		291	293	287	277	270	268	264	272	280	281	283	282
		Edh at Noon	30	33	38	43	43	40	43	40	36	33	31	30
All-Sky Solar Radiation	RadAvg		1307	1562	1837	2015	2034	1884	2052	2003	1813	1610	1413	1268
	RadStd		94	116	84	96	101	115	87	73	102	76	66	82

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+1.58	+1.91	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	+1.58	+1.91	N/A	N/A	N/A	N/A

Nomenclature: See separate page